

yellow typed number · gray formula · green answer · dark green Excel function · bold blue a cell you type in · grids show a blank sheet: always start at A1. Course spelling: NORMDIST / NORMINV / NORMSDIST / NORMSINV (the .DIST / .INV forms are identical).

1. Expected value of scenarios (the mean)

When to use: the question lists possible outcomes and the chance of each (a table, or text with “x% chance”). Also at the end of every tree.
Keywords in the question: *expected value, expected profit, on average, with probability, chance of, fair price, scenarios*
How to do it: list each outcome; check the probabilities add to 1; multiply outcome × probability; add them up.
Example: flats sold 0/1/2/3 with P .1/.3/.3/.3 → 0(.1)+1(.3)+2(.3)+3(.3) = **1.8**. Real estate 145/150/155 with .2/.6/.2 → **150**.
Excel: =SUMPRODUCT(A2:A5, B2:B5)

2. Standard deviation of scenarios (the risk)

When to use: same table as formula 1, when they ask for risk, variability or how much the outcome “swings”. **Always compute the mean FIRST.**
Keywords: *standard deviation, variance, risk, volatility, how much it varies, riskier*
How to do it: for each outcome (outcome – mean)²; multiply by its probability; add up = **variance**; square root = **stdev**.
Example: flats: .1(0–1.8)²+ .3(1–1.8)²+ .3(2–1.8)²+ .3(3–1.8)² = .324+.192+.012+.432 = 0.96 → root = **0.9797**. Forgot the root? The number is huge and in squared units.
Excel: C2: =(A2-\$B\$7)^2 · =SQRT(SUMPRODUCT(C2:C5, B2:B5))
Then Y = a + bX: E = a + b·E(X); stdev = |b|·stdev(X). Y = –50+200X → E 310, sd 195.96. **a** never changes the spread. A tiered payoff is NOT a+bX — rebuild the table on the payoff.

3. Normal: z and probability

When to use: the question says something is normally distributed and gives a mean and stdev. **Forward:** gives a value, asks for a %. **Backward:** gives a %, asks for the value.
Keywords: *normally distributed, probability that, less than, more than, between, at least, percentile, 90% chance, how many to produce, VaR, what score do I need, top 10%*
How to do it: Forward: z = (value – mean)/stdev → NORMSDIST(z) = % below (above: 1 minus; between: larger minus smaller). Backward: % BELOW → NORMSINV → z → value = mean + z·stdev.
Example: grades N(80,10). P(X<90): z=1 → **.8413**. P(X>90) = **.1587**. P(75<X<90) = .8413–.3085 = **.5328**. Top 10%: NORMINV(.90,80,10) = **92.8**. Portfolio N(7,10), worst 5%: NORMINV(.05,7,10) = **–9.45%**.
Excel: =NORMSDIST(z) · =NORMSINV(p) · =NORMDIST(x,μ,σ,TRUE) · =NORMINV(p,μ,σ)

4. SAXBY: combining two variables D = a·X + b·Y

When to use: the question combines TWO variables: two demands, two stocks, revenue from two products, a weighted score, pooling, or compares X with Y. If they are normal, D is normal too and you then use formula 3.
Keywords: *portfolio, invest in each, total demand, combined, correlated, correlation, independent, pooling, more than twice, higher than, exceeds, weighted*
How to do it: 1) set a and b from the question (units per box, dollars invested, weights; comparing X>Y means a=1, b=–1; “twice” means b=–2). 2) mean = a·μX + b·μY (p does not enter). 3) stdev = root(a²σX² + b²σY² + 2abρσXσY). 4) use formula 3.
Example: walk-in N(100,10), phone N(150,15), ρ=.8. Total: mean 250, sd root(100+225+2·10·15·.8) = root(565) = **23.77**. Stock for 99%: NORMINV(.99,250,23.77) = 305.3 → **306**. X–Y: mean –50, sd root(100+225–240) = **9.22**, P(X>Y) = 1–NORMDIST(0,–50,9.22,1) ≈ **0**.
Excel: =B4*B2+C4*C2 · =SQRT(B4^2*B3^2+C4^2*C3^2+2*B4*C4*B5*B3*C3)

5. Trees, “at least one” and Bayes

When to use: several stages, several “ifs”, repeated years/trials, or “given that the test was positive”.
Keywords: *if positive / if negative, survey then drill, over N years, at least one, never, given that, tested positive, false alarm, default*
How to do it: along a path **multiply** (AND); between paths **add** (OR); endpoints add to 1. “At least one” = 1 – P(neither) = 1 – (1–p)^N. Bayes: P(A|+) = path(A and +) ÷ all “+” paths.
Example: AAA p=.0002, 25 yrs: 1–.9998²⁵ = **.004988**. HIV: .006·.999 ÷ (.006·.999 + .994·.01) = **.376**. \$1 → ×1.1 / ×0.9 twice: E **1.04**, sd **.141**.

Excel from scratch: what to type

Type	Meaning
=	every formula starts with =
SUMPRODUCT(A2:A5, B2:B5)	row×row then add: mean of scenarios
SQRT(x) x^2	square root · square
\$B\$7	\$ locks the cell when you drag the formula down
NORMSDIST(z)	z → % BELOW
NORMSINV(p)	% below → z
NORMDIST(x, mean, sd, TRUE)	value → % below, no z needed. Always TRUE
NORMINV(p, mean, sd)	% below → value, directly
1-NORMDIST(. . .)	% ABOVE
AVERAGE / MEDIAN / STDEV	raw data (see p.2 for .S vs .P)
CORREL(A2:A50, B2:B50)	correlation of two columns
SKEW / KURT	shape (KURT is excess: add 3)

Most used z values (area to the LEFT)

80%	90%	95%	97.5%	99%
0.8416	1.2816	1.6449	1.9600	2.3263

Above: use 1 minus. “10% above” = 90% below = z 1.2816. Below the mean: negative z (5% below = –1.6449). ±1σ 68.3% · ±2σ 95.4% · ±3σ 99.7%.

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Session 1 · Dispersion and z

- Standard deviation** = typical distance from the mean. The mean alone is not enough: class A (68 to 72) and class B (40 to 100) both have mean 70; stdevs 1.41 and 21.21. Same units as the data.
- Variance** = mean of squared deviations. We square because raw deviations always cancel to zero. Stdev = root of variance, back to original units.
- Stdev is never negative and is sensitive to outliers. 2, 5, 7, 9, 10: mean 6.6, $\Sigma(x-\bar{x})^2 = 41.2$, $S^2 = 41.2/4 = 10.3$, $S = 3.21$.
- Mean vs median**: the mean is dragged by outliers (Bill Gates in the bar); the median is resistant. Replace 10 by 10000 → median still 7, mean 2004.6.
- STDEV.S** (sample, divides by $n-1$) vs **STDEV.P** (population, divides by n). Lecture 1 handout: "Sample Standard Deviation ... Excel: STDEV.S". **The practice-solution sheet uses STDEV.P — check which the marker expects; the difference is small for large n.**
- Empirical rule (Normal only)**: $\pm 1\sigma$ 68.3%, $\pm 2\sigma$ 95.4%, $\pm 3\sigma$ 99.7%. Weight $\mu 180$ $\sigma 10$ → 95.4% are 160–200.
- z** = (value – mean)/stdev, unitless; compares different things. Exam 90 in $N(75,10)$: $z = 1.5$ → 93rd percentile; in $N(80,10)$: $z = 1$ → 84th. Shanghai 11.75 ($\mu .57$, $\sigma 7.25$) $z = 1.54$ vs NYSE 10.39 ($\mu .46$, $\sigma 4.49$) $z = 2.21$ — the NYSE day was rarer. $|z| < 1$ normal; 2 unusual; > 3 very rare.
- Value from z: value = mean + $z \times$ stdev. Works for ANY distribution; only turning z into a % needs Normality.

Session 2 · Shape and correlation

- Right skew (positive)**: long tail to the RIGHT, **mean > median** (income: mean 106.5k, median 83.7k; movie revenue; market caps). Mode < Median < Mean.
- Left skew (negative)**: tail to the LEFT, **mean < median** (easy exam grades, satisfaction scores). Trick: the name follows the TAIL, not the peak.
- Symmetric: mean = median. **|skew| ≥ 0.5 is significant**. Excel SKEW(); by hand $\approx 3(\text{mean} - \text{median})/\text{stdev}$.
- Kurtosis** = sharp peak AND heavy tails (more extreme values, both directions) than a Normal. Normal = 3 on the lecture scale. **Excel KURT() is EXCESS (Normal = 0): add 3**. Heavy tails mean a Normal approximation understates disasters (S&P 500: March 2020 was a -6σ month).
- Quiz: kurtosis 3 vs 5, same mean/sd → B has more extreme calls both ways; the Normal approximation is riskier for B.
- Correlation ρ** : -1 to 1 ; 0 means no LINEAR relationship (a parabola can have $\rho = 0$). Sign = direction; size = how tightly points hug a line. Corr is **not the slope** (0.30 does not mean 30 cents per dollar) and **not causation** (feet vs education 0.849). Correlations break in crises (LTCM 1998).
- Covariance** = $\rho\sigma_X\sigma_Y$ → $\rho = \text{cov}(\sigma_X\sigma_Y)$. $\rho .5$, $\sigma 10$ and 4 → cov 20.

Session 3 · Probability and expected value

- Random variable**: possible outcomes + a probability for each (must add to 1). Continuous if any value is possible (heights); discrete if countable (units sold).
- E[X]** is a probability-weighted average = long-run average = fair value = balance point. It need not be a possible outcome (a die: 3.5). Always compare E first, then risk.
- Complement**: $P(\text{at least one}) = 1 - P(\text{none})$. AAA default .0002/yr over 25 years: $1 - .9998^{25} = .004988$. Never add up the "at least one" cases — there is only ONE never-path.
- Tree**: multiply along a branch, add across branches, endpoints add to 1. Write each endpoint as net payout (subtract what was invested on that path). Two dice: 36 paths, $E(\text{sum}) = 7$. All casino games have $E < 0$ (chuck-a-luck $E = -\$0.046$).
- Bayes**: split the population by condition FIRST, then by test result; answer = wanted path ÷ all paths with that test result. A 99.9%-accurate test for a 0.6% condition: $P(H|+) = 0.38$. The healthy branch is enormous, so most positives are false alarms.
- Linear transform** $Y = a + bX$: $E = a + bE(X)$, $V = b^2V(X)$, $\text{sd} = |b|\text{sd}(X)$. The constant shifts the mean only.

Session 4 · SAXBY and diversification

- a and b are context multipliers: portfolio weights (0.4 and 0.6), units per box (1 and 2), score weights (0.35 and 0.65), pooling (1 and 1), a difference (1 and -1). They need not add to 1.
- Correlation does NOT change the mean**, only the risk. Changed ρ and the mean did not move? Correct.
- $\rho = +1$: maximum risk, stdev is the simple sum ($a\sigma_X + b\sigma_Y$), no diversification at all. $\rho = 0$: cross term disappears. $\rho = -1$: minimum risk, can reach zero.
- Stdevs do not add** (like Pythagoras): variance first, then the root. Weights enter SQUARED. Use decimals throughout (8% = 0.08) or plain numbers throughout — never mixed.
- Portfolio effect**: a mix can be less risky than EITHER asset. Real estate X ($\sigma 3.16$) and Y ($\sigma 2.68$), $\rho -.5$, 40/60 → $\sigma 1.47$; best mix $w^* = .44$ → $\sigma 1.45$, below both. Idiosyncratic randomness averages out.
- Target return**: $w\mu_X + (1-w)\mu_Y = \text{target}$ → w. 10%/20% assets, target 15% → $w = .5$. **Target risk**: quadratic in w → two roots, keep the one in $[0,1]$ (15%/35%, $\rho .5$, target 20% → $w = .60$). **Min risk**: $w^* = (\sigma_Y^2 - \rho\sigma_X\sigma_Y)/(\sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y)$.
- Can the portfolio beat the better asset's return? NO. Fall below the worse? NO. Have lower risk than both? **YES**.
- Pooling**: two GPU demands $\mu 20$ $\sigma 10$ each, $\rho .2$, cover 3σ : separate 50+50 = 100; pooled $\mu 40$, $\sigma \sqrt{240} = 15.49$ → 86.5. Combining reduces uncertainty; at $\rho = +1$ it would not.
- Efficient frontier** (Markowitz): risk on x, return on y. Lower ρ bends the curve LEFT. The leftmost point is the minimum-variance portfolio.

Session 5 · Normal distribution

- Symmetric bell, defined only by μ and σ . Continuous: **probability = area**; $P(\text{exact value}) = 0$, so $<$ and $\leq$ give the same result.
- Standard normal Z: mean 0, stdev 1. Standardise = turn x into z. NORMDIST and NORMINV always work with the area to the **LEFT**.
- 4 types**: below (NORMDIST), above (1 minus), between (larger minus smaller), inverse (NORMINV: % → value). **Draw the bell and shade first**.
- Demand $N(1200,130)$: $P(<1350) = .8749$; stock for 95%: $\text{NORMINV}(.95,1200,130) = 1413$ ($1200 + 1.645 \times 130$).
- VaR**: portfolio $N(7\%,10\%)$: $\text{NORMINV}(.05,7,10) = -9.45\%$. Sentence: "with 95% confidence the loss in a year will not exceed 9.45%". S&P $N(8.79,15.75)$: $P(<0) = .288$; $P(0-12\%) = .292$; worst 5% = -17.11 .
- Normal + SAXBY**: $aX + bY$ of normals is normal. Do SAXBY, then treat the result as ONE normal. $P(X > Y) = P(X - Y > 0)$: $a=1$, $b=-1$, cut-off 0.
- Gifted & Talented**: .35·SAI + .65·NAI, both $N(100,16)$. Ann 121.6. $\rho=1$ → $\sigma 16$, pct .9115; $\rho=.5$ → $\sigma 14.1$, pct .937. Lower ρ = smaller σ = higher percentile.
- Historical sales do not show unmet demand (censored) — a reason to model demand as Normal.

Descriptive statistics in Excel (sessions 1 and 2)		
What	Type	Reminder
Mean / median	=AVERAGE(A2:A50) / =MEDIAN(A2:A50)	outliers move the mean, not the median
Stdev (raw data)	=STDEV(A2:A50) (=STDEV.S)	handout says .S ($n-1$); .P divides by n
Correlation	=CORREL(A2:A50, B2:B50)	cov = $\rho\sigma_X\sigma_Y$
Skew / kurtosis	=SKEW(A2:A50) / =KURT(A2:A50)	SKEW > 0 right tail; KURT + 3
z of a data point	=(A2 - AVERAGE(A:A))/STDEV(A:A)	$ z > 2$ is rare
Percentile of a point	=NORMSDIST(z)	only if roughly Normal

Sign of the cross term $2 \cdot a \cdot b \cdot \rho \cdot \sigma_X \cdot \sigma_Y$			
a · b	ρ	term	effect on stdev
+ (sum, portfolio)	–	negative	decreases — diversifies
+ (sum)	+	positive	increases
– (difference X–Y)	+	negative	decreases — they move together, so the gap is stable
– (difference)	–	positive	increases
any	0	zero	disappears (independent)

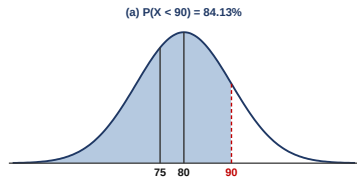
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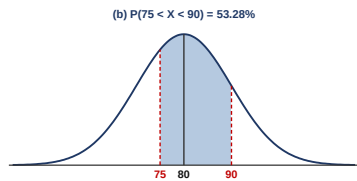
The logic of every Normal question

Question gives the **VALUE** and asks for a % $\rightarrow$ value $\rightarrow$ z $\rightarrow$ **NORMSDIST(z)** = % below. Question gives the % and asks for the value $\rightarrow$ % below $\rightarrow$ **NORMSINV** $\rightarrow$ z $\rightarrow$ value = mean + z·sd. NORMSDIST / NORMSINV / NORMDIST / NORMINV always work with the area to the **LEFT**. Draw the bell, mark μ , mark x, shade what is asked. Shaded right? Then 1 minus.

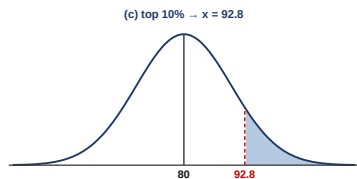
Worked: grades $\mu = 80$, $\sigma = 10$ (Lecture 5)



1. Estimate: above the mean $\rightarrow$ more than 50%.
2. $z = (90 - 80) / 10 = 1.0$.
3. **NORMSDIST(1) = .8413**.



1. Between = larger area minus smaller.
2. $z = 1 \rightarrow .8413$; $z = -5 \rightarrow .3085$.
3. $.8413 - .3085 = .5328$.



1. Top 10% above = **90% below**.
2. **NORMSINV(.90) = 1.2816**.
3. $x = 80 + 1.2816 \times 10 = 92.8$.

(a) .8413 · (b) .5328 · (c) 92.8

B6	fx	
	=NORMDIST(90,B1,B2,TRUE)	
	-	
	NORMDIST(75,B1,B2,TRUE)	
	A	B
1	Mean	80
2	Std dev	10
3	(a) value	90
4	z	1.00
5	P(X<90)	0.8413
6	(b) P(75 to 90)	0.5328
7	(c) z of 90%	1.2816
8	x	92.82

Type in the gray/green cells:

B4 = (B3 - B1) / B2 — z: value minus mean, over sd

B5 = NORMSDIST(B4) — area to the left of z. Same as

=NORMDIST(B3,B1,B2,TRUE)

B6 = NORMDIST(90,B1,B2,TRUE) -

NORMDIST(75,B1,B2,TRUE) —

larger minus smaller

B7 = NORMSINV(0.9) — z that

leaves 90% below (top 10% above)

B8 = B1 + B7 * B2 — mean + z·sd.

Check: =NORMINV(0.9,B1,B2)

Check two ways: (a) also

=NORMDIST(90,B1,B2,TRUE).

(c) also

=NORMINV(0.9,B1,B2). "More

than 90":

=1 - NORMDIST(90,B1,B2,TRUE)

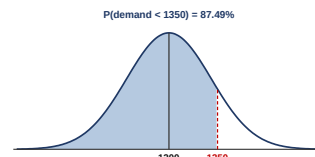
= .1587.

Common mistakes: forgetting TRUE; feeding NORMINV 0.10 instead of 0.90 for "top 10%"; forgetting 1 minus for "more than"; mixing % and decimals.

Variations that may appear (one Normal variable)

If the exam changes to	Type	With $\mu 80$ $\sigma 10$
P(X > 90), "more than", "at least"	=1 - NORMDIST(90, B1, B2, TRUE)	.1587
value with only 5% BELOW (worst 5%, VaR)	=NORMINV(0.05, B1, B2)	63.55 (N(7,10): -9.45)
value of the top 10% / "90th percentile"	=NORMINV(0.90, B1, B2)	92.82
"how many to stock for a 95% chance"	=NORMINV(0.95, mean, sd) $\rightarrow$ round UP	N(1200,130): 1413
68 / 95 / 99.7 (no Excel)	mean $\pm 1\sigma$, $\pm 2\sigma$, $\pm 3\sigma$	70-90 · 60-100 · 50-110
"within k stdevs"	=NORMSDIST(k) - NORMSDIST(-k)	k=2: .9545
which of two outcomes was rarer	z for each, compare z (bigger = rarer)	Shanghai 1.54 vs NYSE 2.21
"what percentile am I"	=NORMSDIST(z) $\rightarrow$ $\times 100$	z 1.5 $\rightarrow$ 93rd
P(X < 0), chance of a loss	=NORMDIST(0, mean, sd, TRUE)	N(8.79,15.75): .288

Worked 2: demand $\mu = 1200$, $\sigma = 130$ and "which was rarer?"



1. $z = (1350 - 1200) / 130 = 1.15 \rightarrow$ NORMSDIST = **.8749**.
2. Stock so that 95% of days are covered: NORMINV(.95,1200,130) = $1200 + 1.645 \times 130 = 1413.8 \rightarrow$ **1414**.
3. Shortest / worst 5%: NORMINV(.05,...) = 986.

B5	fx	=NORMINV(0.95,B1,B2)
	A	B
1	Mean	1200
2	Std dev	130
3	P(<1350)	0.8749
4	P(>1350)	0.1251
5	Stock for 95%	1413.8
6	Order (round up)	1414

B3 = NORMDIST(1350, B1, B2, TRUE) —

area to the left of 1350

B4 = 1 - B3 — above

B5 = NORMINV(0.95, B1, B2) — value

with 95% below

B6 = ROUNDUP(B5, 0) — whole units

Which of two outcomes was more unusual?

1. Each value against ITS OWN mean and sd.
2. $z = (\text{value} - \text{mean}) / \text{sd}$ for each.
3. Bigger |z| = rarer. Ignore the raw values.

B5	fx =(B2-B3)/B4		
	A	B	C
1		Shanghai	NYSE
2	Value	11.75	10.39
3	Mean	0.57	0.46
4	Std dev	7.25	4.49
5	z	1.54	2.21

B5 = (B2 - B3) / B4 — drag right to C5

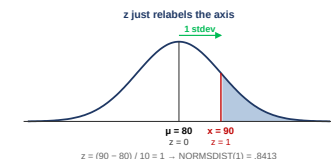
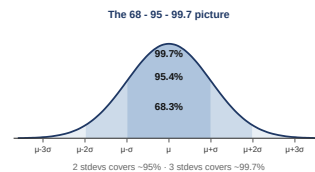
Shanghai +11.75% (z 1.54) vs NYSE +10.39% (z 2.21): the smaller move was the rarer one. z works for ANY distribution; only turning z into a % needs Normality.

Percentile of a score: z =

$(90 - 75) / 10 = 1.5 \rightarrow$

NORMSDIST(1.5) = .933 $\rightarrow$ 93rd percentile. If the class mean were 80: z = 1 $\rightarrow$.841 $\rightarrow$ 84th. The mean and sd are everything.

Empirical rule and the standard-normal picture



Left: the three bands every Normal has. Right: z is just a relabelled axis — the curve does not change. $z = (90 - 80) / 10 = 1 \rightarrow$ NORMSDIST(1) = .8413.

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Scenario table: E(X), V(X), stdev (Lecture 3 contractor)

1. List every outcome with its probability; check they add to 1.
2. Mean B7: $0(.1)+1(.3)+2(.3)+3(.3) = 1.8$.
3. Deviation² C2: $(0-1.8)^2 = 3.24$; drag down.
4. Variance B8: $3.24(.1)+.64(.3)+.04(.3)+1.44(.3) = 0.96$.
5. Std dev B9: $\text{root}(0.96) = 0.9797$.

$$E = 1.8 \cdot \sigma = 0.9797$$

Then profit $Y = -50 + 200X$ (\$K): $E(Y) = -50 + 200 \times 1.8 = 310$; $\text{sd}(Y) = 200 \times 0.9797 = 195.96$. Or type the profits (-50, 150, 350, 550) in column A with the same probabilities.

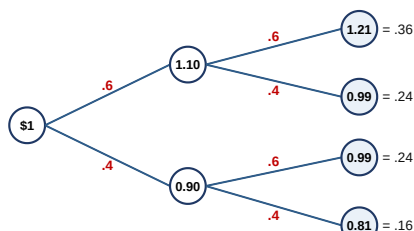
Tiered bonus (0 / 15 / 52.5 / 82.5 with the same P): not a+bX. Put the payoffs in column A → $E = 45$, $\text{sd} = 30.19$.

B9	fx =SQRT(SUMPRODUCT(C2:C5, B2:B5))		
	A	B	C
1	Outcome	Prob	Deviation ²
2	0	0.1	3.24
3	1	0.3	0.64
4	2	0.3	0.04
5	3	0.3	1.44
6	Check ΣP	1.00	
7	Mean	1.80	
8	Variance	0.96	
9	Std dev	0.9797	

Type in the gray/green cells:

B6 =SUM(B2:B5) — must be 1.00 or something is missing
 B7 =SUMPRODUCT(A2:A5, B2:B5) — outcomes × probabilities, added up
 C2 =(A2-B\$7)^2 — outcome minus mean, squared. Drag down to C5
 B8 =SUMPRODUCT(C2:C5, B2:B5) — deviations² × probabilities, added
 B9 =SQRT(B8) — root of the variance

Probability tree and “at least one”



0.99 twice → .24+.24 = .48 · E = 1.04

1. A branch per outcome; the probability goes ON the branch.
2. Along a path multiply: $.6 \times .6 = .36$.
3. The same endpoint reached twice? Add: $.24 + .24 = .48$.
4. Endpoints add to 1; then formula 1 and 2 on the endpoints.

$$E = 1.04 \cdot \sigma = 0.141$$

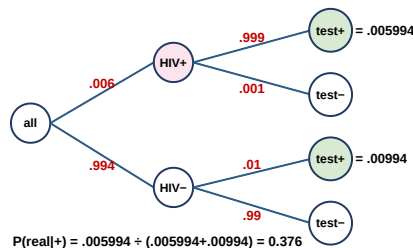
B2	fx =0.6*0.6		
	A	B	C
1	Endpoint	Prob	Value
2	up, up	0.36	1.21
3	up, down	0.24	0.99
4	down, up	0.24	0.99
5	down, down	0.16	0.81
6	Mean	1.0404	

B2 =0.6*0.6 — product of the branch probabilities on that path
 B6 =SUMPRODUCT(B2:B5, C2:C5) — same mean formula as the scenario table

$$P(\text{at least one in } N) = 1 - (1 - p)^N$$

AAA p .0002, 25 years: $=1-(1-0.0002)^{25} = .004988$. “At least one” has ONE never-path — compute that and subtract. Never add up the cases.

Bayes: “tested positive — is it real?”



$$P(\text{real}+) = .005994 + (.005994 + .00994) = 0.376$$

1. Split by **condition** first (.006 / .994), then by **test** result.
2. Multiply along each path.
3. Circle every path that tests positive: .005994 and .00994.
4. Answer = the path you want ÷ all circled paths.

$$P(\text{HIV}+) = 0.376$$

B6	fx =B4/(B4+B5)	
	A	B
1	P(condition)	0.006
2	P(+ condition)	0.999
3	P(+ no condition)	0.01
4	real positives	0.005994
5	false alarms	0.00994
6	P(real +)	0.3762

B4 =B1*B2 — condition AND positive
 B5 =(1-B1)*B3 — healthy AND positive

B6 =B4/(B4+B5) — wanted path over all positive paths

A 99.9%-accurate test on a 0.6% condition is still mostly false alarms. The healthy branch is huge.

Same mean, different risk (Lecture 3 real estate) · a casino game

B6	fx =SQRT(SUMPRODUCT((A2:A4-B5)^2, B2:B4))			
	A	B	C	D
1	x	P(X)	y	P(Y)
2	145	0.2	147	0.4
3	150	0.6	150	0.2
4	155	0.2	153	0.4
5	Mean	150		150
6	Std dev	3.16		2.68

Both houses are worth 150 on average; X swings more (σ 3.16 vs 2.68), so Y is the safer buy. Compare E first, then σ . Risk is measured by σ , not by the chance of a loss alone.

B6	fx =SUMPRODUCT(A2:A5, B2:B5)	
	A	B
1	Profit	Prob
2	-1	0.579
3	1	0.347
4	2	0.069
5	10	0.005
6	Mean	-0.046
7	Std dev	1.288

Chuck-a-luck: $E = -\$0.046$ per \$1 bet — every casino game has $E < 0$. The variance formula in one cell: $=\text{SUMPRODUCT}((A2:A5-B6)^2, B2:B5)$, then SQRT.

Variations that may appear (scenarios and trees)

If the exam changes to	Type	Answer
more scenarios (5 rows)	extend the columns: A2:A6, B2:B6, C2:C6	same formulas
profits given in text (“40 with 25%, -10 with 75%”)	type them as a table first; check $\Sigma P = 1$	E 2.5, sd 21.65
“minimum chance of success to break even” ($E = 0$)	$p \times \text{gain} + (1-p) \times \text{loss} = 0 \rightarrow$ solve p	$40p - 10(1-p) = 0 \rightarrow p = 20\%$
compare two decisions	compute each E; pick the larger; tie → smaller sd	
a survey / test costs money before the decision	tree; subtract the cost on EVERY path	
probability of one specific path	multiply its branches: $=0.5*0.5$	25%
$P(A \text{ or } B) / P(A \text{ and } B)$	$P(A)+P(B)-P(A \text{ and } B)$; independent: $P(A) \times P(B)$	
exactly k successes in N	$=\text{BINOMDIST}(k, N, p, \text{FALSE})$	at most k: TRUE
payout = fixed + per-unit × X	$E = a + b \cdot E(X)$; $\text{sd} = b \cdot \text{sd}(X)$	$-50+200X$: 310, 195.96

yellow typed number · gray formula · green answer · dark green Excel function · bold blue a cell you type in · grids show a blank sheet: always start at A1. Course spelling:

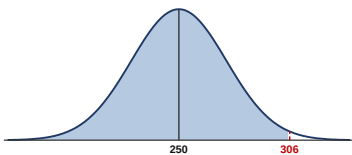
NORMDIST / NORMINV / NORMSDIST / NORMSINV (the .DIST / .INV forms are identical).

The reusable SAXBY block (Lecture 5, Ex 3: walk-in + phone demand)

- Right variable: total demand $D = 1 \cdot X + 1 \cdot Y \rightarrow a = 1, b = 1$.
- Mean: $100 + 150 = 250$. (ρ does not enter.)
- Std dev: $1^2 \times 100 + 1^2 \times 225 = 325$; cross term $2 \times 1 \times .8 \times 10 \times 15 = +240$; $\text{root}(565) = 23.77$.
- Stock for 99% of days: $\text{NORMINV}(.99, 250, 23.77) = 305.3 \rightarrow 306$.
- Do not add the two 99% levels separately. Combine first, percentile after.

D ~ N(250, 23.77) · stock 306

99% below the stock level



B7	f _x	=SQRT(B4^2*B3^2+C4^2*C3^2+2*B4*C4*B5*B3*C3)		
	A	B	C	
1		X	Y	
2	Mean	100	150	
3	Std dev	10	15	
4	Weight (a, b)	1	1	
5	Correlation	0.8		
6	Mean D	250.00		
7	Std dev D	23.77		
8	Stock (99%)	305.30		

Type in the gray/green cells:

B6 =B4*B2+C4*C2 — mean of D: weight × mean X + weight × mean Y
B7 =SQRT(B4^2*B3^2+C4^2*C3^2+2*B4*C4*B5*B3*C3) — SAXBY: $a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y$, all under the root
B8 =NORMINV(0.99, B6, B7) — mean + z(99%) × sd. Round UP → 306

This A1:C8 block is the same for every two-variable question. Build it once; change only the yellow numbers (means, stdevs, weights, correlation), then add the NORMDIST / NORMINV line you need.

Weighted score (Gifted & Talented)

- Score = .35·SAI + .65·NAI, both N(100,16) → $a=.35, b=.65$.
- Mean 100 either way. $\rho=1$: $\text{sd} = 16$. $\rho=.5$: $\text{sd} = \text{root}(31.36+108.16+2 \cdot .35 \cdot .65 \cdot .5 \cdot 256) = 14.1$.
- Ann 121.6 → $\rho=1$: $\text{NORMDIST}(121.6, 100, 16, 1) = .9115$; $\rho=.5$: $.937$.

Lower correlation → smaller sd → the same score is a HIGHER percentile.

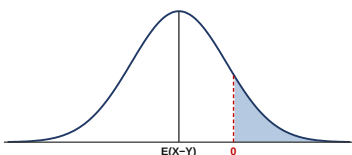
B8	f _x	=NORMDIST(121.6, B6, B7, TRUE)		
	A	B	C	
2	Mean	100	100	
3	Std dev	16	16	
4	Weight (a, b)	0.35	0.65	
5	Correlation	0.5		
7	Std dev D	14.10		
8	P(<121.6)	0.937		

Variations that may appear (SAXBY)

If the exam changes to	Type	With the demand numbers
chance total demand exceeds 280	=1 - NORMDIST(280, B6, B7, TRUE)	10.3%
95% service level instead of 99%	=NORMINV(0.95, B6, B7)	289.1 → 290
X and Y independent	B5 = 0 (cross term vanishes)	sd $\text{root}(325) = 18.03$
2 units of X per order, 3 of Y	B4 = 2, C4 = 3	mean 650
"X more than twice Y"	B4 = 1, C4 = -2, then P(D>0)	
dollars invested instead of weights	a = \$ in X, b = \$ in Y; means are RETURNS; result in \$	P5a: \$10K/\$10K → \$8,000, \$4,183
weights as % of the portfolio	a = .5, b = .5 → result in return %	P5: mean 40%, sd 20.9%
pooling: is the shared sd smaller?	compare B7 with B3 + C3	23.77 < 25 yes
they give covariance, not ρ	$\rho = \text{cov} / (\sigma_X\sigma_Y)$ first	

P(X > Y) — move everything to one side

P(X - Y > 0): shade RIGHT of zero



- $X > Y$ becomes $X - Y > 0 \rightarrow a = 1, b = -1$. ("more than twice Y" → $b = -2$.)
- Mean: $100 - 150 = -50$.
- Std dev: $100 + 225 - 240 = 85 \rightarrow \text{root} = 9.22$ (b^2 kills the sign; the cross term flips).
- $P(D > 0) = 1 - \text{NORMDIST}(0, -50, 9.22, \text{TRUE}) \approx 0$: zero is 5.4σ above the mean.

P(walk-ins > phone) ≈ 0

B8	f _x	=1 - NORMDIST(0, B6, B7, TRUE)		
	A	B	C	
1		X	Y	
2	Mean	100	150	
3	Std dev	10	15	
4	Weight (a, b)	1	-1	
5	Correlation	0.8		
6	Mean D	-50.00		
7	Std dev D	9.22		
8	P(D>0)	0.0000		

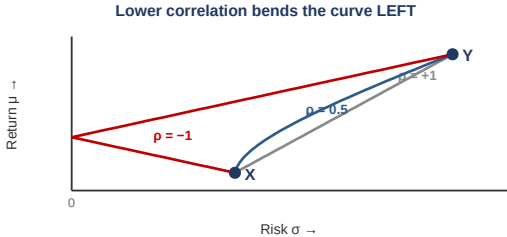
C4 -1 — the minus sign is the whole trick
B8 =1 - NORMDIST(0, B6, B7, TRUE)
 — 1 minus the area below 0, with mean B6 and sd B7

Mark **ZERO** on the D curve, not the mean. Independent ($\rho = 0$)? The cross term simply disappears.

Practice P5c (DSL 50%/45% vs CableM 30%/25%, $\rho = -.4$, RETURNS not dollars): $a=1, b=-1 \rightarrow$ mean 20%, sd $\text{root}(.2025 + .0625 + .09) = 59.58\%$ → $P(D>0) = 63.1\%$.

yellow typed number · gray formula · green answer · dark green Excel function · bold blue a cell you type in · grids show a blank sheet: always start at A1. Course spelling: NORMDIST / NORMINV / NORMSDIST / NORMSINV (the .DIST / .INV forms are identical).

Portfolio P = w·X + (1-w)·Y



$E(P) = w \cdot \mu_X + (1-w) \cdot \mu_Y$	ρ	frontier
$\sigma(P) = \sqrt{(w^2\sigma_X^2 + (1-w)^2\sigma_Y^2 + 2w(1-w)\rho\sigma_X\sigma_Y)}$	+1	straight line, no benefit
	0	bends left
	-1	touches zero risk

① Target return (linear): $w = (\text{target} - \mu_Y) / (\mu_X - \mu_Y)$. 10%/20%, target 15% → $w = .5$. 15%/25%, target 16% → $w = .9$.
② Target risk (quadratic, two roots): $A = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$; $B = 2\rho\sigma_X\sigma_Y - 2\sigma_Y^2$; $C = \sigma_Y^2 - \text{target}^2$; $w = (-B \pm \sqrt{B^2 - 4AC}) / 2A$, keep the root in $[0,1]$. σ 15%/35%, ρ .5, target 20% → $w = .60$.

③ Minimum risk: $w^* = (\sigma_Y^2 - \rho\sigma_X\sigma_Y) / (\sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y)$. Real estate σ 3.16/2.68, ρ -.5 → $w^* = .44$, min $\sigma = 1.45$ (below both). w^* outside $[0,1]$ = needs short selling → answer with the boundary.

Can the portfolio...	
return more than the better asset?	NO
return less than the worse asset?	NO
have lower stdev than BOTH?	YES

Leftmost point = minimum-variance portfolio. Pooling (GPUs μ_{20} σ_{10} each, ρ .2, cover 3 σ): separate 50+50 = 100; pooled 40 + 3×15.49 = 86.5. The means add; the sd shrinks.

Exam checklist

- 1. Write in one sentence what is asked: below, above, between, or find the value?
- 2. Scenarios? Probabilities add to 100%, each profit on the row of its probability.
- 3. Two variables? a and b from the question (units, dollars, weights; 1 and -1 to compare; -2 for "twice"). Build the A1:C7 block.
- 4. "More than / exceed / higher": 1 minus. NORMSINV and NORMINV: always the area BELOW.
- 5. Estimate first (above or below the mean? more or less than 50%), then check two ways (z + NORMSDIST, and NORMDIST directly).
- 6. μ and σ on the same scale; keep all decimals; round only the final answer; units/stock round UP.
- 7. KURT + 3. STDEV.S per the handout. Did you take the square root?
- 8. Show your process in Canvas: formula, numbers, result with units.

Review session: understand the question in words before calculating; estimate the size and direction; check units; solve two ways; keep all decimals.

Practice midterm answers (for checking)

P	Answers
1	a TRUE (median of 7 = 4th value) · b FALSE (right tail: mean > median) · c FALSE (slope ≠ ρ)
2	Uris Petrol: E \$2.5M, σ \$21.65M; with survey E \$2.5M (tie)
3	Call centre N(240,40): P(<180) 6.68% · P(180–300) 86.64% · top 1% at 333.05 s
4	Lego A N(50K,10K), B N(30K,7K): 68K fully met 96.41% · ρ -.1, 90%: 130.98K · ρ =.5 → more · P(A>2B) 28.05%
5	DSL/CableM \$10K each, ρ -.4: E \$8,000, σ \$4,183 · P(CD < 5%) 4.71% · P(DSL > CableM) 63.1%

Standard normal P(Z < z)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Negative z: P(Z < -z) = 1 - P(Z < z).

Question → tool

If you see	Use	Lecture
a list of raw numbers; "describe", "skew"	AVERAGE / MEDIAN / STDEV / SKEW / KURT+3	L1
"which was more unusual", "percentile"	z each, compare z ; NORMSDIST	L1
"correlation", two columns, true/false about ρ	CORREL; sign, not slope, not cause	L2
scenario table, "expected", "standard deviation"	mean and sd of scenarios (formulas 1, 2)	L3
"profit = fixed + per unit"	$a + bE(X)$, b ·sd	L3
"survey", "if positive", several ifs, over N years	tree → mean; $1-(1-p)^N$	L3
"given the test was positive"	Bayes	L3
"normally distributed", "less / more / between"	value → z → %	L5
"find x", "90% chance", "how many to produce", VaR	% → z → value (NORMINV)	L5
two variables, weights, "portfolio", "correlated"	SAXBY → Normal	L4, L5
"A more than twice B", "X higher than Y"	$P(X - bY > 0)$, SAXBY	L5
"target return / target risk / safest mix"	portfolio formulas ①②③	L4
"pooling", shared facility, "k stdevs of cover"	SAXBY $a=b=1$; $\mu + k\sigma$	L4